

SPHERE: a Novel Approach to 3d and Active Sound Localization

A naturalistic approach to spatial hearing studies

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ABSTRACT

In everyday life, localizing a sound source entails more than the sole extraction of auditory cues to define its three-dimensions: spatial hearing also takes into account the available visual information (e.g. cues to sound position) and resolves perceptual ambiguities through active listening behavior (e.g. exploring the auditory environment with head movements). We introduce a novel approach to sound localization in 3D named SPHERE which exploits a commercially available Virtual Reality Head-mounted display system with real-time kinematic tracking to combine all of these elements (controlled positioning of a real sound source, recording of participants' responses in 3D, controlled visual stimulations and active listening behavior). SPHERE allows accurate sampling of 3D spatial hearing abilities, and allows detection and quantification of the contribution of active listening.

CCS CONCEPTS

• **B7; Applied computing** → Life and medical sciences; Systems biology.

KEYWORDS

Spatial hearing, virtual reality, head-movements, motion tracking, active perception

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1 INTRODUCTION

Spatial hearing is a fundamental ability and is three-dimensional, multisensory and active: 1) we estimate azimuth, elevation and distance of sounds, 2) perceive the visual context in which they occur, and 3) most importantly, spatial hearing is an active process; we explore the auditory environment with head and body movements to resolve perceptual ambiguities in sound localization. When spatial hearing abilities are investigated, however, most of these naturalistic aspects of spatial hearing are considerably constrained (limited set of sound locations, response limited to one or two dimensions, full vision of the loudspeakers providing visual priors, constrain head-movements to ensure reproducibility of sound characteristics). In sum, currently dominant approaches to sound localization typically do not capture the sound localization behavior during real-life naturalistic conditions. To overcome some of these limitations, some studies exploited the advantages of virtual reality for both visual and auditory stimuli [Majdak et al., 2011]. Nonetheless, it remains quite a challenge to track and update the 3D virtual position of sounds in real-time as a function of 3D head-movements [Lida, 2019], and the transfer to clinical settings remained rather limited in particular for participants with hearing aids or cochlear implants.

We report on a novel approach that aimed to circumvent most of these limitations using real sounds coupled with visual virtual reality solutions. To examine the potentials and the validity of this approach, we studied sound localization in 20 normal hearing individuals leveraging, as a benchmark, the well-established advantage of moving the head while listening [Wallach, 1940]. We compared static (head still) vs. active listening condition (head free to move) for sounds localization.

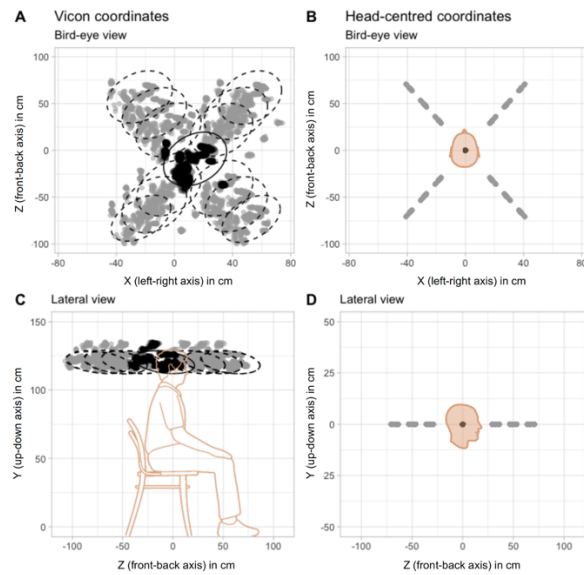


Figure 1: A to D: From Vicon to subject head-centered coordinates. Bird-eye and lateral view of initial head position (in black) and 12 pre-determined speaker locations (in gray) for all participants, in VICON reference frame (A-C) and head-centered coordinates (B-D).

2 DESIGN AND PROCEDURE

We took advantage of current VR technology to determine a method to guide manually a real loudspeaker to precisely pre-determined head-centered coordinates in each trial. Critically, this head-centered placement of the loudspeaker, coupled with criteria for sound emission based on the participant's head and eye posture measured in real-time (Oculus Rift Development Kit 2 system incorporating an eye-tracking system + Vicon motion capture system 2.0) allowed to test sound localization abilities without imposing a physical restrain on the participant's posture and allowed sound locations to be replicated among trials and participants. Furthermore, tracking head and hand motion in real-time (Vicon system) allowed to assess the contribution of head-movements to sound localization and full description of the kinematic of hand-pointing responses to 3D sound source position. Participants were told to indicate the sound position under two conditions: 'static listening' (head still throughout sound presentation) and 'active listening' (free to move the head to search for the sound). We tested 12 pre-determined sound locations. The position of all tracked elements recorded in Vicon reference frame (loudspeaker, head center and direction, hand) was re-computed in head-center reference frame, and kinematic analysis of head and hand data were performed.

3 RESULTS

Our results provide validation for the SPHERE system and paradigm: (1) when referenced to the center of the head, computer-controlled

placement of the speaker led to an error below 1 cm across all target positions, all participants and all recording sessions (mean differences between pre-determined and actual speaker location $x = 0.87 \pm 0.01$ cm, $y = 1.15 \pm 0.02$ cm, $z = 0.93 \pm 0.01$ cm) (see Figure 1); (2) we replicate the contribution of head-movements to sound localization, i.e. significant improvement of sound localization performance in active compared to static condition, in azimuth (active 15.9 deg.; static 26.1 deg.) and elevation (active 21.2 deg.; static 31.4 deg.) for front sound sources.

4 DISCUSSION AND CONCLUSION

Our system is effective for controlling the delivery of sounds in 3D while measuring in real-time participant behavior, without imposing a physically constrained posture and with minimal training required. Our system is sensitive to capture the head-behavior effect on sound localization, improving specifically azimuth and elevation perception. If several promising research avenues are possible with our approach [Valzolgher et al., 2020; Coudert et al., 2022], certain limits could be resolved with technical adjustments: collect responses beyond reaching space need to adapt the response method [Valzolgher et al., 2022]; the HMD constitutes a physical obstacle to sound diffusion, reducing its size will have less impact.

Finally, this approach paves the way for future research, clinical and industrial applications that will leverage the full potential offered by having embedded a VR HMD in the SPHERE system [Salemme et al., 2021].

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